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Optimized Stimulation Efficiency with Degradable Particulates in High-Anisotropy Carbonate Reservoirs: Comparative Study of Horizontal and Vertical Wells in Iraq

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Abstract

Highly anisotropic carbonates require unique stimulation fluid placement methodologies and one of the ways to improve this is by utilizing degradable particulate material, which will be the focus of this manuscript. The primary objective was to help improve stimulation efficiency and reduce the number of stages, overall treatment time, and requirement for prolonged workover rig operations. A comparative evaluation between horizontal and vertical well geometries was also conducted to assess particulate placement behavior and diversion effectiveness.

Degradable particulates were introduced as a temporary diversion agent during acid stimulation treatments in a carbonate reservoir with high permeability contrast. The particulate blend was engineered to fully degrade under downhole conditions to eliminate post-treatment intervention and across multiple field applications on both vertical and horizontal wells. Real-time data from pressure response, surface rates, and fluid distribution confirmed diversion performance. Post-treatment production data and wellbore imaging tools confirmed fluid coverage and zonal activation.

The degradable-particulate diversion workflow delivered marked improvements in stimulation efficiency for both vertical and horizontal high-anisotropy carbonate wells. After a baseline matrix-acid schedule, particulates were injected between stages to temporarily seal treated zones and force acid into bypassed intervals at matrix stimulation rates. Real-time surface pressure records rose from 15% to 60% after each diversion stage, followed by a pressure drop with the 15% HCl, confirming improved fluid penetration homogeneity and effective zonal isolation. Post-operation tests showed uplift in stabilized oil rates and an observed increase in average wellhead pressure versus pretreatment values, which demonstrated the direct impact of improved fluid placement and reservoir contact.

Encouraged by these gains, a transition occurred from complex ported work strings that struggle to apply the technology to simple open-end tubing with bridge plugs, which streamlined operations and extended diversion reach in longer intervals. This optimized design cut workover rig time by 25%, reduced stage count by 40%, and removed the requirement for straddle packers. Field-wide rollout sustained production

increases of 25 to 35% for the treated wells, which established degradable particulates as a cost-effective, low-footprint solution that helped improve stimulation coverage in complex carbonate reservoirs.

How degradable particulate diversion can simplify stimulation workflows and maximize treatment efficiency in complex carbonate reservoirs is highlighted. It also provides valuable insights into particulate behavior in various wellbore geometries, which helps improve the design of future stimulation strategies in anisotropic formations.

Introduction

Carbonate reservoirs rarely appear uniform because they are built by living organisms in dynamic marine settings and then altered by diagenesis and tectonics. This process produces highly irregular porosity-permeability distributions that make stimulation and fluid placement difficult. Carbonate reservoirs formed from marine organisms (i.e., corals, shells, algae) create distinct skeletal structures and pore systems. Diagenesis further alters these pore systems through cementation, which fills pores in certain areas and leaves others open, or through dissolution by acidic fluid (i.e., meteoric water, organic acids), which selectively dissolves carbonate materials to create vugs, molds, and channels.

Sedimentation and tectonic movement also affect carbonate reservoirs. The sedimentation rate of carbonate rock is relatively high compared to sandstone reservoir systems because the carbonate reservoir originated at the site of deposition with limited transport distance compared to sandstone. However, this condition varies by region. Iraq and Arabian carbonate reservoirs are well known for their heterogeneity because of these combined factors.

This heterogeneity and anisotropy strongly influence the economic performance of each well. They determine the volume of oil in place, represented by porosity, and the transportable oil, represented by permeability. Based on reservoir characteristics, production could decrease or increase accordingly (Malik et al. 2018).

In terms of stimulation, each well requires periodic acid treatment based on productivity. However, proper stimulation design shall be implemented to target the entire interval, which represents the reservoir thickness or pay zone. Pumping acid alone rarely proves sufficient because the thief zones within heterogenous layers absorb the acid, which leaves large sections untreated and results in production rates below expectations.

Diversion is required to redirect acid volumes from one zone to another based on interval flow intake, which depends on skin, permeability, and pressure. Diversion methods fall into two main categories (Van Domelen 2017):

- Mechanical: Seals the formation with rigid components, such as packers, ball sealers, particulates, lost circulation material (LCM), and fibers
- Chemical: Relies on fluid mobility control (viscosimeters), differential pressure methods (foam), etc.

In high anisotropy zones with long intervals, chemical methods often fail unless pumped at very high volumes, which significantly increases cost. These zones require near wellbore (NWB) bridging, perforation blocking, or isolation of each zone. Chemical diversion is generally straightforward, but mechanical diversion involves greater complexities and considerations. For example, particulates, LCM, and fibers cannot be pumped through small slots, such as perforated joints deployed with a work string, or they might only be pumped in limited concentrations, which often reduces their effectiveness (Fig. 1).

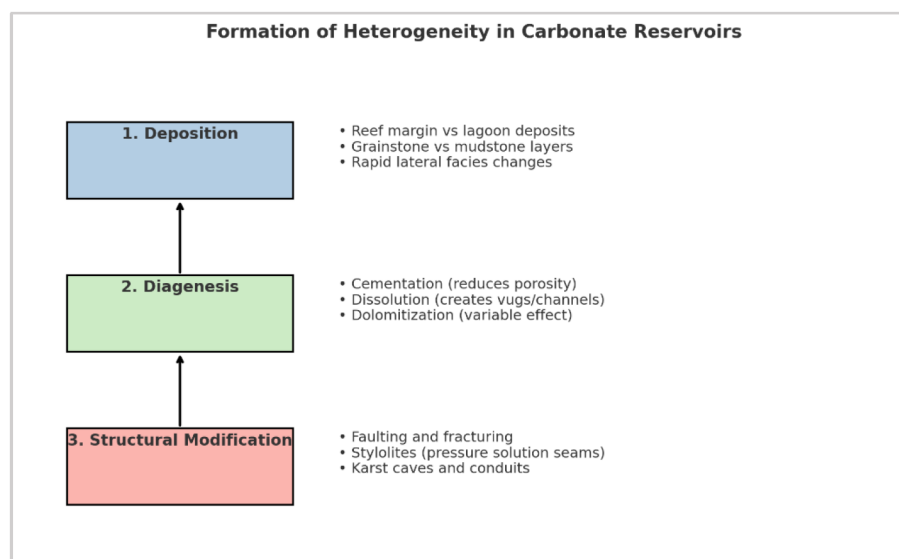


Figure 1—General carbonate reservoir heterogeneity formations.

The method applied in high-anisotropy reservoirs used degradable particulates, known as "degradable particulates near wellbore" (DPNWB), designed for NWB bridging under matrix stimulation conditions. The evaluation of this diversion method's results, selected after prior use of other diverters (i.e., straddle packers, chemical diverters) in the same reservoir section of various wells, is discussed.

Particulate Diverter Classification

Particulates serve two main functions—diversion from high-permeability zones to low-permeability zones and a temporary LCM that blocks the formation for a period of time based on their degradation profile. Several diverter types are available within the market, classified by solubility, degradation characteristics, and particle size, each tailored to specific reservoir conditions (Sierra et al. 2017).

Diversion increases the efficiency of acid stimulation through flow direction to achieve full coverage of the perforated reservoir thickness. Some particulates also support hydraulic acid fracturing, although their characteristics in that operation differ from those in a matrix stimulation, primarily because of particle size requirements.

In general, particulate diverters fall into the following categories (Medina et al. 2015) (Table 1):

- **Degradable Particulates:** Polylactic acid (PLA) is a thermoplastic polyester derived from renewable sources, such as corn starch or sugarcane. PLA hydrolyzes in water and produces lactic acid that is soluble and can help slightly dissolve carbonate rock. Other types include PLA granules, polyglycolic acid (PGA), poly(lactic-co-glycolic) acid (PLGA), and polycaprolactone (PCL). It breaks by temperature—the higher the temperature, the faster the hydrolysis—pH drops (because acidic environments accelerate degradation), and the smaller the particle size, the faster it degrades.
- **Soluble Particulates:** Salts serve this role and are considered crystalline solids that dissolve in formation water after diversion completes. Magnesium chloride dissolves slower than other salts and can be carried in a non-dissolving material until the acid reacts and water production initiates dissolution.
- **Insoluble Particulates:** Examples include ceramic, sand, bauxite granules, and glass beads. These materials remain intact until they are mechanically removed or flushed out and are commonly used during hydraulic fracturing.
- **Composite/Self-Assembling Systems:** This is a combination of several materials, such as fibers, particulates, and binding agents. This complex system bridges the pore throats or fractures with

fibers, and the particulates fill the fiber mat gaps. Viscoelastic fluid or crosslinkers (binding agents) temporarily stabilize the plug materials in place. Such systems provide high stability in naturally fractured reservoirs with turbulent stimulation flow.

Table 1—Particulate classification and comparison by application.

Type	Material Examples	Removal Mechanism	Typical Applications
Degradable	PLA, PGA, PLGA, PCL	Hydrolysis → organic acids	Carbonate acidizing, restimulation
Soluble	NH ₄ Cl, KCl, NaCl, MgCl ₂	Dissolution in brine	Matrix treatments, openhole completion
Insoluble	Ceramic beads, sand, bauxite	Mechanical removal, backflow	Fracturing diversion
Composite/Self-Assembling	Fiber + particle blends, swellable microspheres	Combination (degrade + dissolve)	High heterogeneity reservoirs, wide PSD

Case studies focused on degradable particulates and demonstrated the technique's effectiveness for temporary shutoff of water losses and the diversion of acid systems in Iraq oilfield locations.

Operation Design

The design of DPNWB operations requires careful consideration of the following:

- **Limitations of Usage:** Application in slotted or perforated work strings often causes slot blockage, which results in operational failure. To reduce this risk, solids concentration is an important variable to consider. The material requires a high-viscosity suspension fluid, which makes it difficult to pump using coiled tubing where particulate concentration is limited.
- **Particulate Loading:** Interval heterogeneity and length govern the required concentration per diversion cycle when using open-ended work strings. Reduced concentration poses challenges as it creates far-field diversion versus NWB injection reduction. Excess concentration results in complete blockage of the interval, screen out, and stimulation failure that requires cleanout.
- **Temperature Limitations:** Each diversion system operates within a defined temperature envelope that should simulate the reservoir treating temperature, minimum cooldown temperature, and formation heat recovery. DPNWB has a functional envelope in wells suitable for medium- to high-temperature reservoirs. During injection, the fluid cools the formation, but heat recovery timing dictates formation stabilization and degradation rate. Reservoirs with lower temperatures profiles could require an external degradation accelerator.
- **Degradability Ratio:** Temperature within the operating range controls breakdown rate. Higher temperature accelerates degradation, while external fluid can further accelerate breakdown.
- **Zone Length:** Long intervals require several diversion stages combined with the anisotropy of the interval.
- **Carrier Fluid:** Minimum volume and maximum concentration provide the most effective coverage of NWB and perforation phases. Suspension capability of the carrier fluid determines the allowable pump rate (based on decant tests).
- **Surface Equipment Setup:** Proper equipment selection plays a vital role in operational success to guarantee operational efficiency and avoid surface equipment blockage or failures.
- **Injection Validation:** Each operation requires validation of injection feasibility before application. Low injection rates could cause particulate settlement and operational failure.
- **Cement Behind the Casing:** Poor cement quality behind the casing allows particulates to invade the voids between the casing and formation. Excess material should be included in the diversion cycles when poor bonds are observed.

With these design considerations addressed, actual well designs and observed particulate behavior can be analyzed (Fig. 2).

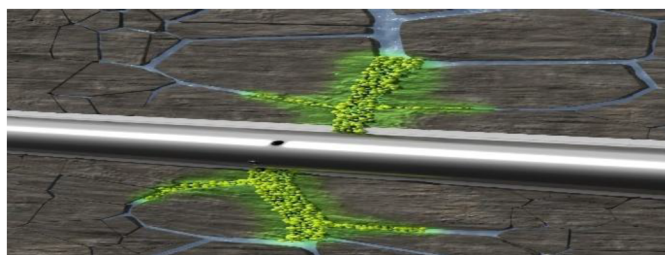


Figure 2—DNWB bridging a NWB area.

At each operation, the term "inflow" refers to the combined effects of permeability, zone length, and skin damage. This parameter governs the flow at each interval of the target layer.

This operation originated as an initiative to reduce rig time from three to five days with straddle packers and workover-based zone-by-zone stimulations to a single operational day with a customized pump schedule that stimulates all intervals in one stage. The initiative reduced the amount of water, brine, and acid to pump in the formation while also achieving broader reservoir coverage.

Well A (First Operation in the Campaign)

Operation Design. Well A is a vertical injector well with three distinct layers. One layer received treatment with DPNWB diversion technology through an open-ended work string and packer, while the other layers relied on straddle packers for stimulation. This well served as a trial. The focus remains on the DPNWB design and outcome, with no evaluation of the other layers. The work string is equipped with downhole pressure and temperature memory sensors (Fig. 3).

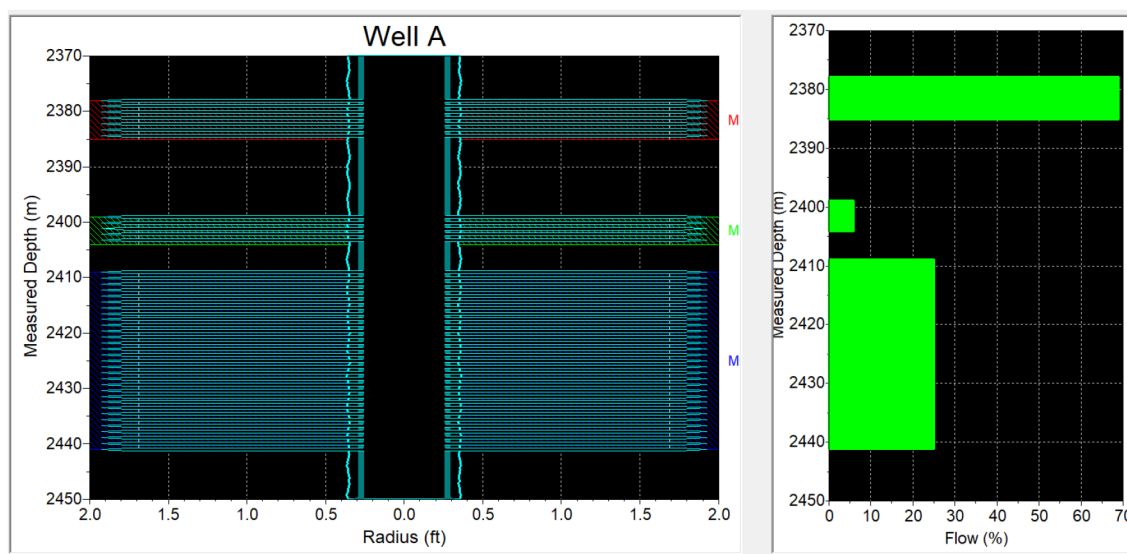


Figure 3—Well A DPNWB Treated Layer perforated intervals with flow percentage per interval to illustrate high contrast governed by skin and permeability [flow based on the injection logging tool (ILT)].

A pickling stage for the work string is mandatory to remove dope, rust, mill scale, and debris attached to the string before fluid injection into the formation. After the pickling stage, an injection test confirms the maximum achievable rate before stimulation of the formation, and it provides a slight diagnostic of the fracture coefficient. The stimulation operation then proceeds in bullheading mode.

Note: For high-rate injection operations, acid and brine formulations require friction reducers to minimize pressure against surface and downhole equipment (Fig. 4 and Fig. 5).

Stage		8.00	LIQ. VOL. (bbl)	LIQ. DENS. (gr/cc)	LIQ. DENS. (lb/gal)	LIQ Rate (bpm)	Cum Vol (bbls)	TIME (min)	Cum Time (min)	Circulation Valve Status
1	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		118.4	1.19	9.91	3.0	118.4	39.5	39.5	Squeeze Position
2	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		2.00	1.19	9.91	1.0	2.0	2.0	2.0	Squeeze Position
3	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		4.00	1.19	9.91	2.0	6.0	2.0	4.0	Squeeze Position
4	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		8.00	1.19	9.91	4.0	14.0	2.0	6.0	Squeeze Position
5	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		12.00	1.19	9.91	6.0	26.0	2.0	8.0	Squeeze Position
6	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		16.00	1.19	9.91	8.0	42.0	2.0	10.0	Squeeze Position
7	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		20.00	1.19	9.91	10.0	62.0	2.0	12.0	Squeeze Position
8	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		24.00	1.19	9.91	12.0	86.0	2.0	14.0	Squeeze Position
9	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		28.00	1.19	9.91	14.0	114.0	2.0	16.0	Squeeze Position
10	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		32.00	1.19	9.91	16.0	146.0	2.0	18.0	Squeeze Position
11	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		33.00	1.19	9.91	16.5	179.0	2.0	20.0	Squeeze Position
12	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		34.00	1.19	9.91	17.0	213.0	2.0	22.0	Squeeze Position
13	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		35.00	1.19	9.91	17.5	248.0	2.0	24.0	Squeeze Position
14	Rig Brine + 0.2% Friction Reducer + 0.05% Surf		36.00	1.19	9.91	18.0	182.0	2.0	26.0	Squeeze Position
15	Shut in	-	-	-	-	-	-	30.0	56.0	Squeeze Position
Total Liquids:			402.4							
									56 min	
									0.93 hrs	

Figure 4—Injection test design stage; the rig brine (KCl or NaCl) includes a friction reducer and a surface tension reducer. The injection test shall not start until only one fluid remains in the string, which helps ensure data accuracy during interpretation.

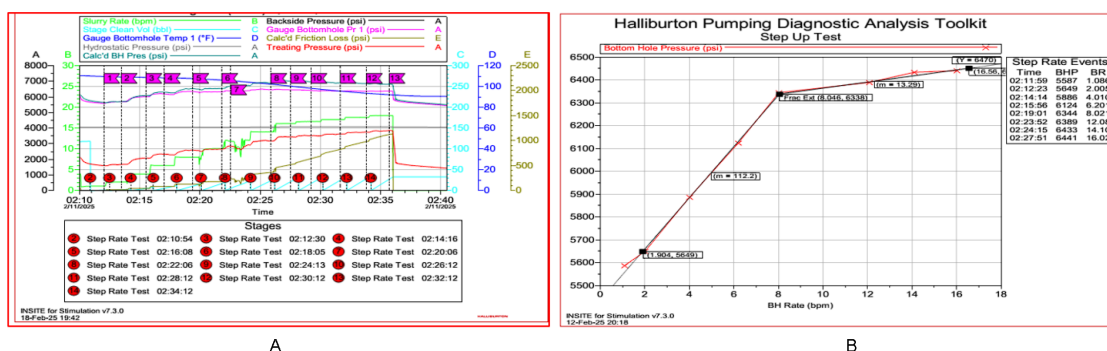


Figure 5—(A) Injection test execution; (B) injection test analysis shows a fracture extension rate of 8 bbl/min and a bottomhole fracture extension pressure of 6,338-psi.

Based on SRT, The fprmation has initiate a fracture extension due to pressure and rate which represented as frac extension rate and frac extension pressure, the test used to determine the maximum allowable surface pressure which was set during the stimulation operation to help prevent fracturing of the formation (Figs. 6 through 8).

Parameter	Value	Unit
Injection Pressure at Max Rate	3854	psi
Max Rate	18.00	bpm
Top Perfs TVD	2378	m
Hydrostatic Pressure (1.19 SG Brine)	4019	psi
ISIP	1764	psi
BH ISIP	5783	psi
FG	0.81	psi/ft
P Total	2090	psi
P pipe	1560	psi
P entry	530	psi
BH Fracture Extension Rate	8	bpm
BH Fracture Extension Pressure	6338	psi

MASP Calculation		
Fluid	Description	Value Unit
	Fracture Gradient	0.81 psi/ft
	Top Perfs TVD	2378 m
	BH Frac Pressure	6338 psi

Hydrostatic Pressure to Top Perf		
Fluid 1	1.19 SG Brine (9.91 ppg)	4019 psi
Fluid 2	15% HCL (8.96 ppg)	3634 psi

MASP		
Fluid 1	1.19 SG Brine (9.91 ppg)	2319 psi
Fluid 2	15% HCL (8.96 ppg)	2704 psi

MASP after applied SF (200 psi)		
Fluid 1	1.19 SG Brine (9.91 ppg)	2119 psi
Fluid 2	15% HCL (8.96 ppg)	2504 psi

Figure 6—Summary table; the outcome of the injection test shows a fracture gradient of 0.81 psi/ft. Based on this value, the maximum pressure during stimulation was set with a safety factor and with hydrostatic adjustments for various fluid.

Stage	Description	Coverage Gal/ft	Liq Rate (bpm)
1	ESS	5.820	8.0
2	15% HCl + 0.2% FR	19.204	8.0
3	Spacer	1.455	8.0
4	DPNWB (2 lb/gal)	0.960	8.0
5	Spacer	3.492	8.0
6	15% HCl + 0.2% FR	13.676	8.0
7	Spacer	3.492	8.0
8	DPNWB (2 lb/gal)	0.698	8.0
9	Spacer	1.455	8.0
10	15% HCl + 0.2% FR	43.646	8.0
11	Spacer	1.455	8.0
12	DPNWB (2 lb/gal)	2.939	8.0
13	Spacer	1.455	8.0
14	15% HCl + 0.2% FR	43.646	8.0
15	Brine 0.05% Surf + 0.2% FR	101.841	Max rate

Figure 7—Designed pumping table; the design includes DPNWB at each of three cycles, with small volumes pumped at matrix stimulation rates identified during earlier evaluations (below frac extension point). Utilized 120 gal/ft acid dosage.

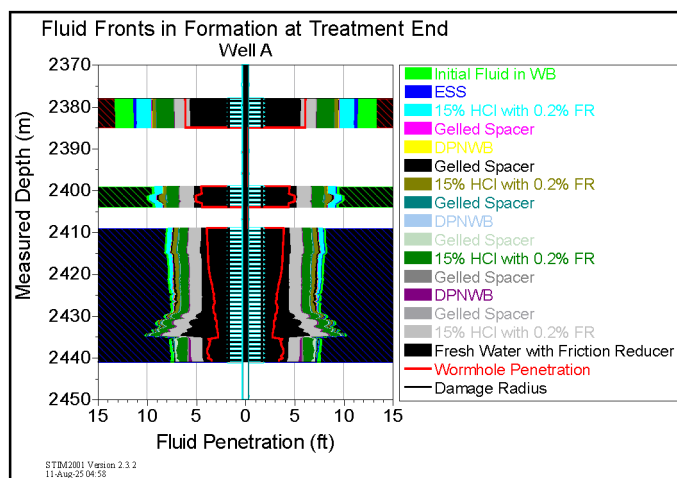


Figure 8—Simulated Design: The simulation addresses wormhole penetration. The flow distribution appears nearly equal despite the first zone with 70% as indicated by the ILT.

Operation Execution. During execution, the equipment setup included a blender and several pump package considerations to overcome challenges with section alignment and discharge line placement (Fig. 9).

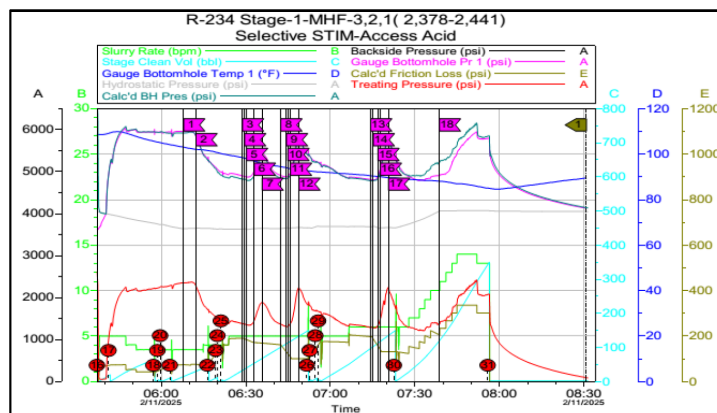


Figure 9—Pressure dropped significantly at flag two, which represented removal of high skin from the entire high inflow zones. Three pressure spikes at downhole contacts with DPNWB confirmed effective diversion. The maximum pressure differential and injection rate (MAPDIR) technique in the final stage achieved additional diversion by rate control.

Execution quality matched design quality, which resulted in excellent post-stimulation injection results. Downhole gauges (DHG) recorded pressure spikes that aligned with the wellhead pressure (WHP), which confirmed accuracy of the friction calculation software based on the 0.2% friction reducer concentration in the fluid system. Downhole temperature (DHT) trends displayed distinct declines, which indicated that new zones entered production flow. Instantaneous shut-in pressure (ISIP) trends deviated from the step-rate test (SRT) values, which confirmed that new zones opened and lower friction was observed (Table 2, Table 3, and Fig. 10).

Table 2—Summary data of Well A stimulation.

Data in Bottomhole Gauges	Well A
Depth (m)	2378 to 2441
Maximum injection rate (SRT) (bbl/min)	18
Maximum injection pressure (SRT) (psi)	3,854 to 6,374
Bottomhole fracture external pressure (psi)	6,338
Bottomhole fracture external rate (bbl/min)	8
Fracture gradient (psi/ft)	0.81
15% HCl acid ΔP surface (psi)	2,348 to 1,350
DPNWB Cycle 1 ΔP bottomhole (psi)	577
DPNWB Cycle 2 ΔP bottomhole (psi)	954
DPNWB Cycle 3 ΔP bottomhole (psi)	955
Pre-injection (B/D)	1,205 (ILT)
Post injection (B/D)	5,750 (sonar test)

Table 3—Timeline of convention stimulation with a straddle packer tool per stage.

Rig Activities Per Stage	Time Estimate (hr)
Setting the packer	1
PT packer	1
Stimulation operation	4 to 12
Unsetting the packer	1
Run in hole or pulled out of hole to the target depth	1
Average total time per stage	10 to 16

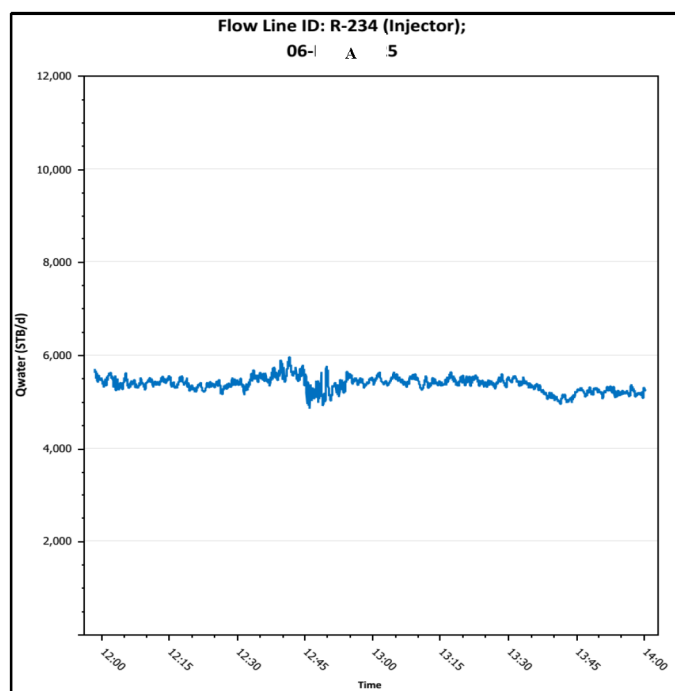


Figure 10—Sonar test on the pipeline with multiple calculations produced post-acid stimulation water injection rates used as comparison with pre stimulation injection test.

Conventional stimulation with a straddle packer that was previously used as a mechanical diversion technique required 16 hours per stage $\times$ 4 stages = 64 hours, plus 48 hours contingency for packer replacement, which resulted in 112 total hours. In contrast, simulation of the entire zone with a single DPNWB stage required only 12 to 20 hours.

Well A underwent stimulation of the entire zone with one DPNWB stage, followed by three straddle packer stages that consumed 100 total hours. Use of the DPNWB reduced wasted time in a busy workover schedule and provided a practical solution to lengthy operations.

Well B (Deviated Well)

Operation Design. Well B drilled with 70° deviation and a long perforated section. The well, a producer that ceased flow, required stimulation. Static reservoir temperature measured 74°C. Previous stimulation attempts using coiled tubing followed by high-rate stimulations showed no significant skin reduction because flow ceased after a short period.

Recent production logging tool (PLT) results indicated that the middle zone delivers the most production, while the other zones provided minimal participation and remained almost fully plugged by skin. The total interval exhibited tight characteristics, with permeability that ranged from 1.8 to 7.5 mD. Total interval thickness (h) measured 152 m; 55 m accounted for the majority of oil contribution. A new perforated interval was planned to allow treatment of the full section with DPNWB technology in a single stage (Fig. 11 and Fig. 12).

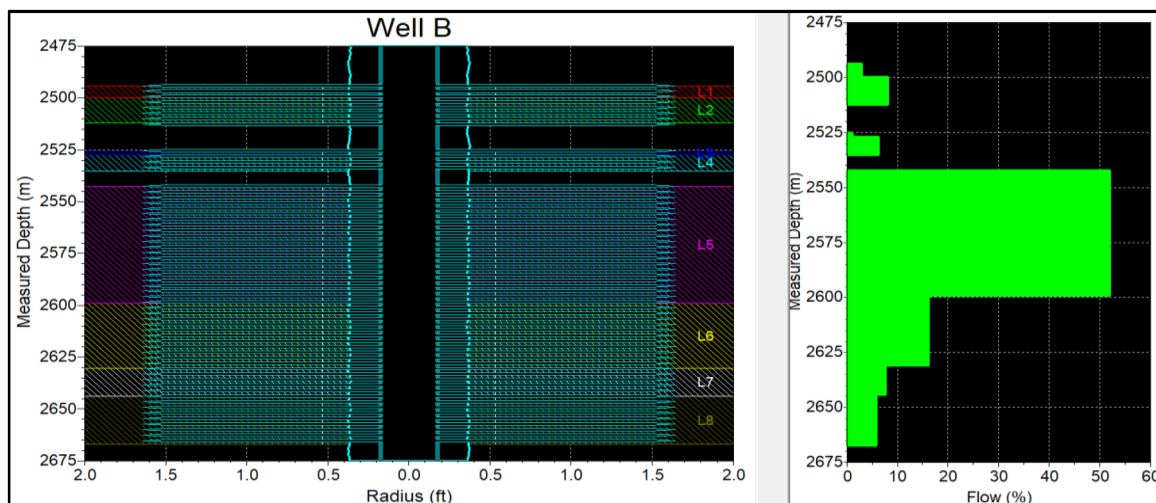


Figure 11—Well B perforated intervals and flow percentages per interval show high contrast based on skin and permeability effects (flow values correspond to PLT results).

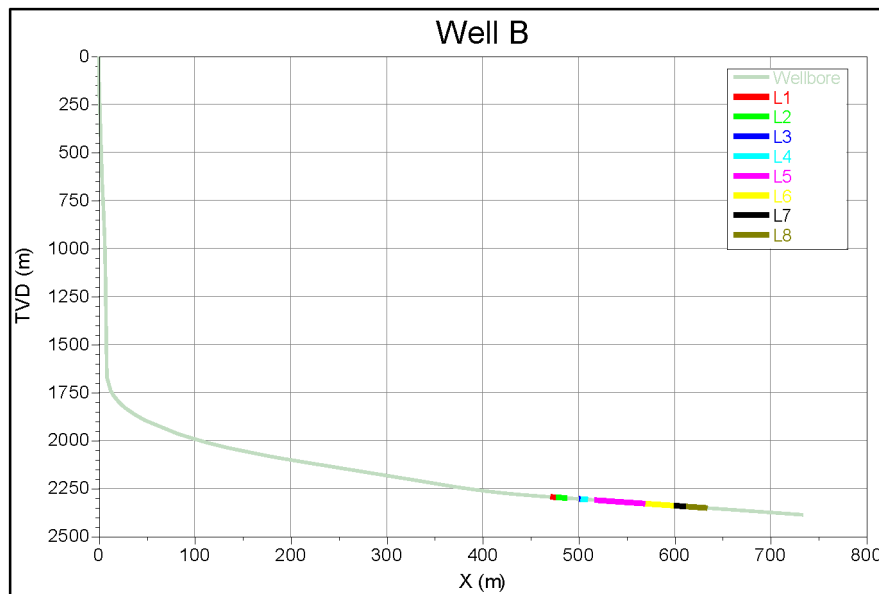


Figure 12—Well B shows the deviation of the wellbore and the layers of the target zone.

Previous stimulation operations with CT and bullheading at high rates without diversion created a vuggy zone because of a high dissolution rate at a specific interval (theoretical but valid). In a deviated wellbore and horizontal section, high tortuosity appears in the NWB because of the long penetrated section of the reservoir. The low static reservoir temperature requires an accelerator to achieve full degradation in ± 3 days.

Similar to Well A, the plan includes pickling, an injection test, and stimulation, but in this case with one stage only. Well B also contains organic deposits that require an increase in mutual solvent to strip the oil phase, which could coat the rock and limit acid reaction with the formation (Fig. 13 and Fig. 14).

Stage	Description	Coverage Gal/ft	Liq Rate (bpm)
1	ESS	0.935	8.0
2	15% HCl + 0.2% FR + 6% MS	14.392	8.0
3	Spacer	0.467	8.0
4	DPNWB (2.5 lb/gal)	1.121	8.0
5	Spacer	0.467	8.0
6	15% HCl + 0.2% FR + 6% MS	18.036	8.0
7	Spacer	0.467	8.0
8	DPNWB (2.5 lb/gal)	0.935	8.0
9	Spacer	0.467	8.0
10	15% HCl + 0.2% FR + 6% MS	14.392	8.0
11	Spacer	0.467	8.0
12	DPNWB (2.5 lb/gal)	0.748	8.0
13	Spacer	0.467	8.0
14	15% HCl + 0.2% FR + 6% MS	10.840	8.0
15	Spacer	0.467	8.0
16	DPNWB (2.5 lb/gal)	0.467	8.0
17	Spacer	0.467	8.0
18	15% HCl + 0.2% FR + 6% MS	7.196	8.0
19	Spacer	0.467	8.0
20	DPNWB (2.5 lb/gal)	0.374	8.0
21	Spacer	0.467	8.0
22	15% HCl + 0.2% FR + 6% MS	7.196	8.0
23	Brine + FR+ Surf	28.036	Max

Figure 13—Designed pump table that shows DPNWB at each of five cycles with small volumes pumped at matrix stimulation rates (identified recently).

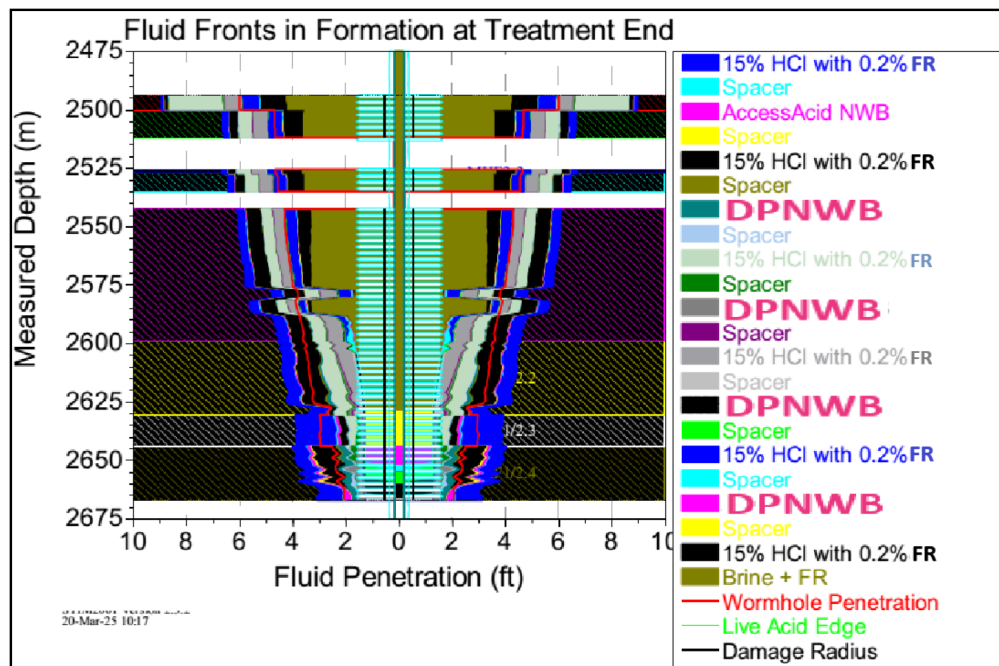


Figure 14—Simulated design addresses wormhole penetration. Flow percentage did not penetrate 50% into the middle zone. The fluid colors appear uneven, which represents good diversion.

Operation Execution. The equipment setup follows the same considerations as Well A (Fig. 15 and Table 4).

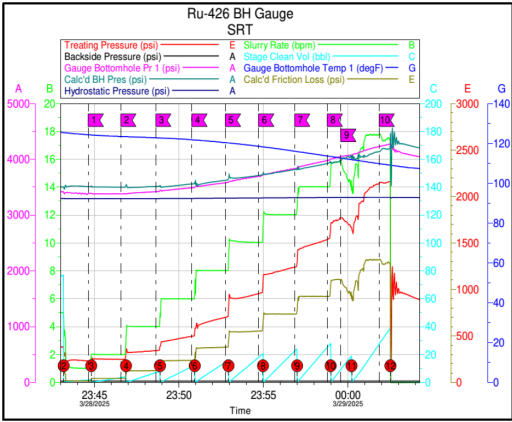


Figure 15—Injection test operation chart shows that pressure never stabilized at each stage. This indicates multiple slight restrictions. High total friction appears on the WHP sensor but not on the downhole gauge.

Table 4—Overall injection test results.

Parameter	Value	Unit
Injection Pressure at Max Rate	2,160	psi
Max Rate	18	bpm
Top Perfs TVD	2,295	m
Hydrostatic Pressure (1.01 SG Brine)	3,289	psi
ISIP	1,012	psi
BH ISIP	4,301	psi
FG	0.57	psi/ft
P Total	1,148	psi
P pipe	1,050	psi
P entry	98	psi
BH Fracture Extension Rate	12	bpm
BH Fracture Extension Pressure	3,839	psi

Pressure reduction from ISIP to zero required an extended period, which indicates poor connection between the reservoir and wellbore (Fig. 16).

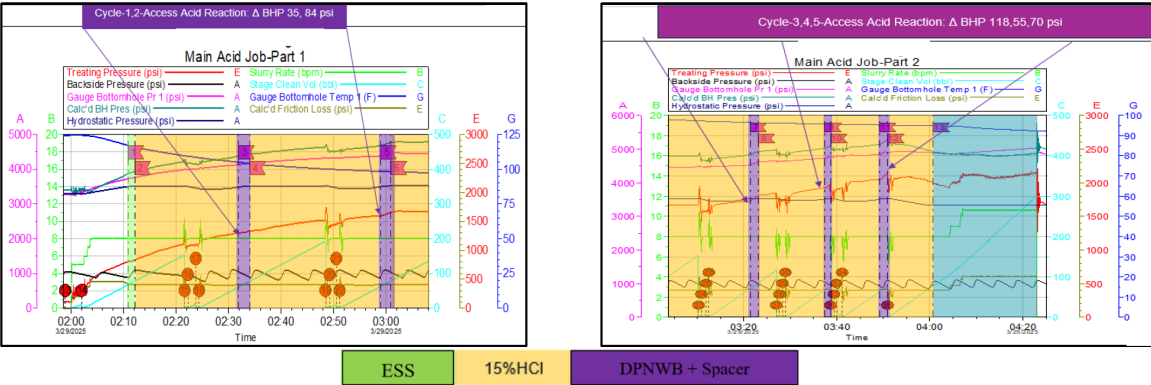


Figure 16—Stimulation operation chart with DPNWB shows various reactions after diverter contact. Slight pressure increases at multiple values, which indicates successful diversion. The severe water hammer observed at end of the treatment means high reservoir-wellbore contact.

The injection test provides insight into the stimulation operation. Most pores or perforations appear plugged.

During the first Cycle of DPNWB, acid contact with the formation produced only minimal reaction. This indicates limited diversion and a low stimulation effect. The homogeneous permeability in the interval created no significant permeability contrast to allow effective acid diversion.

In the second access acid cycle, the access acid concentration was increased. Blender equipment mixed the DPNWB on the fly and modified the concentration based on results from the previous cycle. There was an increase in Delta P (ΔP), and this pressure response continued to rise through the subsequent access acid cycles. The data confirms improved diversion and broader coverage.

During the final stage, the acid reacted actively with the formation. This indicates that the acid successfully reached new zones with current skin damage, which improved overall stimulation efficiency in the interval.

The total time for the stimulation operation on Well Ru-426, from pressure tests of the surface line to rig down after the final stage, was approximately 12 hours. In comparison, a conventional straddle packer stimulation with workover runs would take more than 77 hours. Additionally, the particulates served as LCM, which reduced brine consumption during trips and lowered well control hazards (Table 5).

Table 5—Summary of the data from the Well B stimulation.

Data in Bottomhole Gauges	Well B
Depth (m)	2494 to 2667
Maximum injection rate (SRT) (bbl/min)	18
Maximum injection pressure (SRT) (psi)	2,160 to 6,374
Bottomhole fracture external pressure (psi)	3,839
Bottomhole fracture external rate (bbl/min)	12
Fracture gradient (psi/ft)	0.57
15% HCl acid ΔP surface (psi)	2,100 to 1,800
DPNWB Cycle 1 ΔP bottomhole (psi)	35
DPNWB Cycle 2 ΔP bottomhole (psi)	84
DPNWB Cycle 3 ΔP bottomhole (psi)	118
DPNWB Cycle 4 ΔP bottomhole (psi)	55
DPNWB Cycle 5 ΔP bottomhole (psi)	70

Well C (Work String Tool Advancement)

Operation Design. Well C is a vertical producer well perforated at several layers at the same heterogeneous reservoir. The well ceased flow and remained shut-in for an extended period. Most of the oil originates from the top layer, while three deeper layers remain inactive. PLT results confirm water invasion at the bottom layer.

The plan requires a combined mechanical diversion system that integrates a bridge plug, packer, and DPNWB systems. The retrievable bridge plug (RBP) will isolate the bottom water-bearing zone to reduce stimulation in that interval to zero. DPNWB with acid at matrix stimulation rates will treat the productive oil layers. This design allows all challenges to be resolved in a single run with particulate assistance. The approach reduces rig time compared with straddle packer operations, which often face difficulty achieving consistent diversion (with other diversion systems).

An additional advantage of this system appears when perforations at the top layer produce unwanted gas and perforations at the bottom layer produce unwanted water. In this case, the packer isolates the top layer, the RBP isolates the bottom layer, and the middle layers remain open for stimulation (Fig. 17 and Fig. 18).

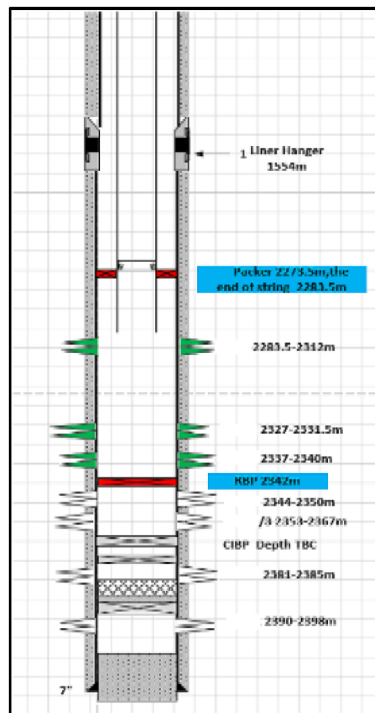


Figure 17—Well sketch shows the unwanted water layer isolated by RBP and the remainder of the casing isolated by the packer to allow stimulation of the gold zones with DPNWB.

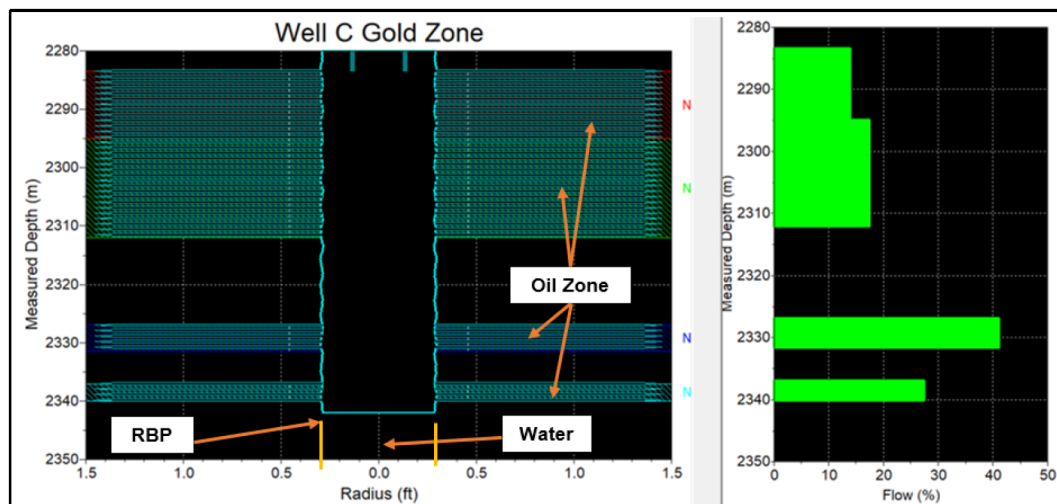


Figure 18—Gold zone flow characteristics reveal that the bottom perforation provides the most oil production despite lower permeability than the top perforated zone, a result of high skin damage that requires removal.

The low-contribution interval at the top is long in terms of diversion and stimulation; therefore, it should be diverted after stimulation. Flow characterization, original permeability, and interval length determine the number of diversion cycles. Flow primarily enters the bottom interval; however, this interval still requires stimulation. This interval defines the first cycle. The top zone contains two heterogeneous zones in terms of permeability; however, both show similar kin because neither produce any fluid. This defines the second cycle of diversion. Stimulation should occur before diversion to treat the first zone and block it, which

allows stimulation of the second zone. Similar to Well A, the sequence requires pickling, an injection test, and stimulation, but this time for only a single stage (Fig. 19).

Stage	Description	Coverage Gal/ft	Liq Rate (bpm)
1	15% HCl + 0.2% FR + 8% MS	24.895	5.0
2	Spacer	3.556	5.0
3	DPNWB (2.0 lb/gal)	2.134	5.0
4	Spacer	3.556	5.0
5	15% HCl + 0.2% FR + 8% MS	59.391	5.0
6	Spacer	3.556	5.0
7	DPNWB (2.0 lb/gal)	6.401	5.0
8	Spacer	3.556	5.0
9	15% HCl + 0.2% FR + 8% MS	35.564	5.0
10	Brine + 0.2% FR+ 0.05% Surf	88.909	Max

Figure 19—Stimulation schedule based on well heterogeneity.

The table defines the acid volume for each stage. The first stage of acid targets the zones already flowing but without negative skin. The subsequent stages require significantly higher volumes to reach and treat the new zones (Fig. 20).

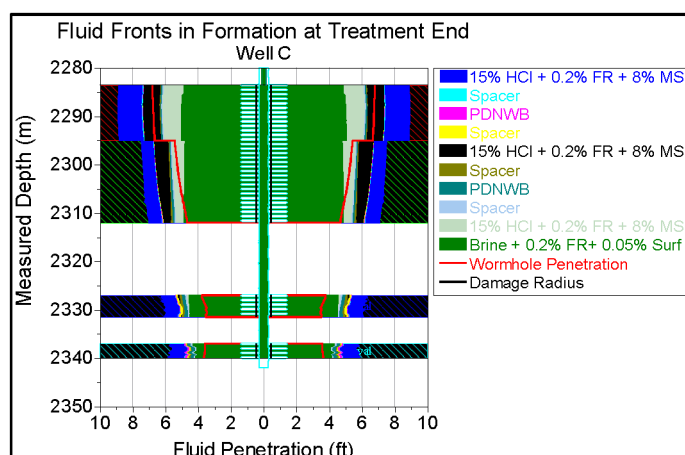


Figure 20—Simulated design pump schedule illustrates wormhole penetration.

Operation Execution. Equipment setup follows the same considerations as wells A and B, with an additional RBP included at this well that requires a run-and-set procedure. The injection test graph (Fig. 21) did not provide an evident interpretation of the fracture gradient because the well has only two short intervals with previous skin damage that restricts the interaction between the wellbore and reservoir. In such cases, even if the injection test undergoes interpretation, the result does not yield accurate data. Another indicator of poor wellbore-to-reservoir connectivity is the absence of water hammer at the ISIP point.

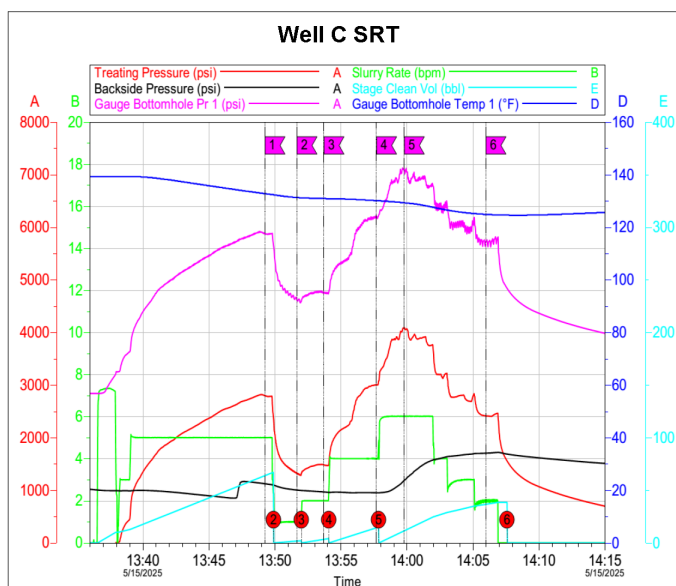


Figure 21—Injection test graph that indicates the maximum rate at maximum pressure.

This injection test provides insight into the stimulation operation. The results show that most pores and perforations are plugged, and, as a result, stimulation cannot proceed at high rates (Fig. 22).

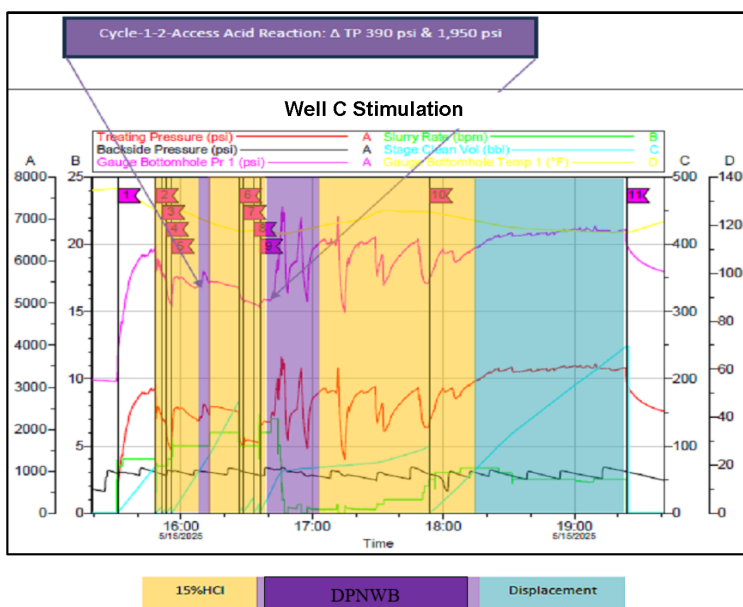


Figure 22—Stimulation operation at matrix stimulation rates with DPNWB.

Acid injection reduced Wellhead pressure from 3,000 to 1,400 psi. This indicates the severe damage that blocked the formation and shows that part of the top perforation opened to flow because of the rate and fluid dynamics. The fluid invaded the top zones and gradually reached the bottom zones. This type of reaction does not relate entirely to the bottom zones intended for stimulation first but represents a mix of flow from both top and bottom zones.

The first cycle contacted the formation and increased the Delta bottomhole pressure (Δ BHP) to 390 psi. This value should have been higher because the plan aimed to block the two bottom zones that had previously contributed flow. However, skin damage reduction at the top zone during the first cycle left open flow paths.

The second cycle contacted the formation and increased the Δ BHP to 1,950 psi, which forced a rate reduction to the minimum applicable value. This response indicated that the entire formation was blocked and that extra particulates settled on the RBP because of the low rate. Suspension of fluid supported particulates during this stage. Once acid reconnected with the reservoir, pressure dropped, which indicated that the pill was severe and required reduction (e.g., from 18 to 12 bbl). This also confirmed deep acid penetration and extra wormhole development.

During displacement, the rate could not exceed 2.5 bbl/min because of high blockage in the NWB, which was represented by high pressure. It is recommended to maintain the DPNWB slurry volume at the minimum, as applicable. An additional challenge appeared during retrieval of the RBP back to surface because the particulates settled on the fish neck. The obstruction was successfully circulated with the work string, and the RBP was retrieved to surface without issue (Table 6 and Fig. 23).

Table 6—Stimulation operation summary.

Data in Bottomhole Gauges	Stage 1
Depth (m)	2283.5 to 2340
Maximum injection rate (SRT) (bbl/min)	6
Maximum injection pressure (SRT) (psi)	4,090
Bottomhole fracture external pressure (psi)	N/A
Bottom hole fracture external rate (bbl/min)	N/A
Fracture gradient (psi/ft)	N/A
15% HCl acid Δ P surface (psi)	3,040 to 1,430
Access acid Cycle 1 Δ P bottomhole (psi)	390
Access acid Cycle 2 Δ P bottomhole (psi)	1,590
Preproduction (B/D)	1,000 of oil + 100 of water (PLT)
Post production (B/D)	1,561 of oil + 343 of water (sonar test)

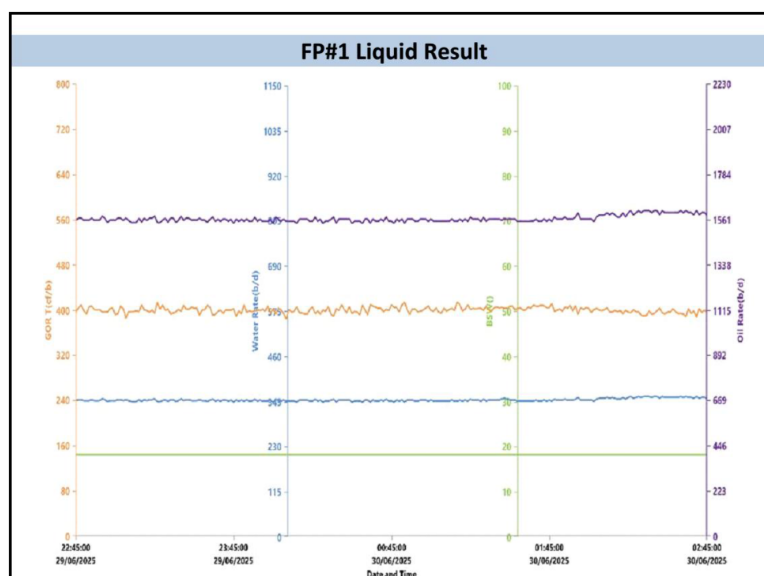


Figure 23—Sonar test on the pipeline with multiple calculations managed to get the rates of water injected into the well after acid stimulation.

Oxidizers cause only limited degradation at low concentrations. High temperatures (65° to 130°C) produce rapid, near-complete dissolution. At 85°C, 3% NaOH achieved approximately 74% dissolution in 24 hours and approximately 89% in 48 hours. At 65°C, dissolution reached approximately 73.5% in 24 hours and approximately 89% in 144 hours (Fig. 25).

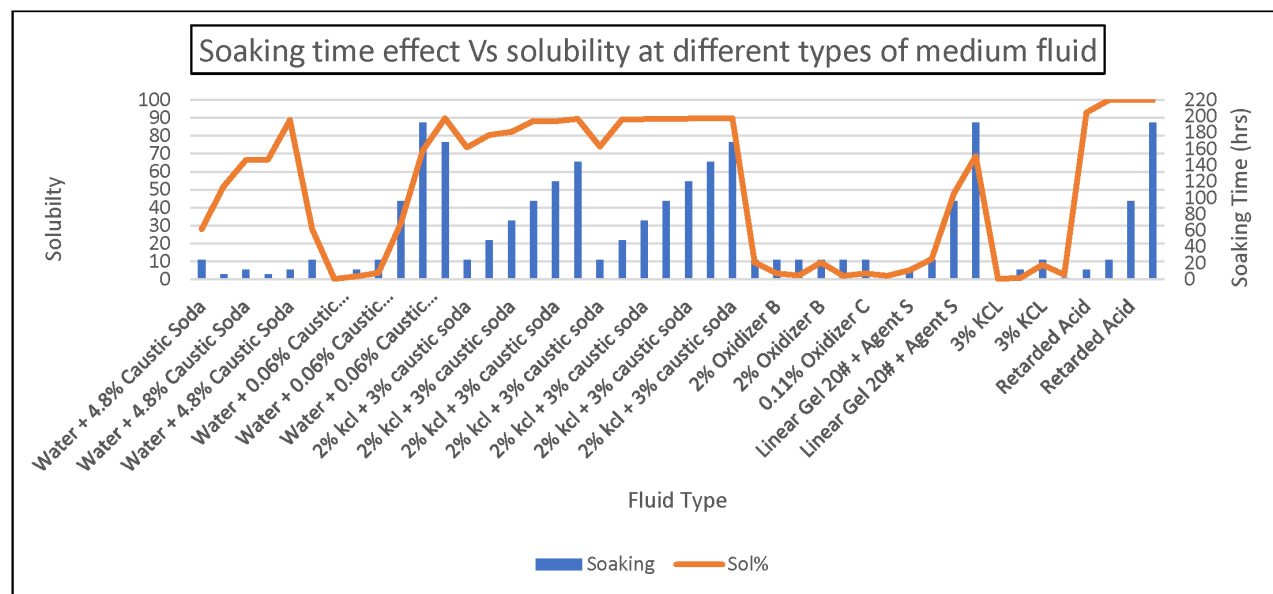


Figure 25—Effect of soak time on particulate degradation soaked in various fluid.

Particulates that require longer exposure to dissolution solutions degrade in an exponential trend. NaOH (caustic soda) solutions exhibited an evident trend: higher concentration produces higher solubility at 24 hours. Fig. 24 and Fig. 25 demonstrate that time and temperature can be projected on each other.

Decant Test (Free-Fall Test)

This test measures the time required for the fluid to suspend the particulates that fall freely inside. The results provide insight into flow rates during execution and provide visibility on carrier fluid performance. The decant test is performed as follows:

1. Mix 500 mL of carrier fluid.
2. Fill a 250 mL cylinder with the carrier fluid.
3. Measure the fluid height in the cylinder with a ruler or measurement tape.
4. Drop the particulates onto the carrier fluid and record the time until the first particle touches the cylinder bottom.
5. Use the recorded time to calculate settling velocity:

$$\text{Setting Velocity} = \text{Length or Height (ft)} / \text{Decant Time (min)} \quad \text{Eq. 1}$$

Generally, decant time should range from 10 to 50 minutes. Less than 10 minutes means the particulates cannot sustain shape during pumping at medium or low rates. More than 50 minutes can increase carrier fluid viscosity to a level unsuitable for pumping.

Conclusion

The field deployment of degradable particulate diversion technology has demonstrated its effectiveness in improving fluid placement and zonal coverage in highly anisotropic carbonate reservoirs. By providing temporary yet reliable diversion, the particulates enabled more uniform acid distribution across both vertical

and horizontal wellbores, leading to measurable increases in production and reservoir contact. Operationally, the optimized workflow reduced stage count and rig time while eliminating the need for complex straddle packer systems, confirming the value of degradable particulates as a cost-efficient, low-risk alternative to conventional diversion methods.

These results reinforce the broader stimulation strategy previously outlined in our SPE work, where innovative diversion approaches were shown to enhance matrix acidizing efficiency. The present study extends that foundation by offering comparative insights into particulate behavior in different well geometries and by demonstrating how degradable particulates can simplify treatment design without compromising effectiveness.

Overall, degradable particulate diversion provides operators with a practical tool to maximize reservoir contact, streamline stimulation execution, and achieve sustained production improvement in complex carbonate environments. Its proven performance highlights the importance of integrating degradable materials into future stimulation workflows as the industry continues to seek efficient, reliable, and intervention-free diversion solutions.

Nomenclature

B/D	barrel per day
CBL	cement bond log
Δ BHP	delta (differential) bottomhole pressure
DPNWB	degradable particulates near wellbore
ESS	emulsified solvent system
ISIP	instantaneous shut-in pressure
ILT	injection logging tool
lbm/gal	pound per gallon
LCM	loss circulation material
MAPDIR	maximum pressure differential and injection rate
NWB	near wellbore
PCL	polycaprolactone
PGA	polyglycolic acid
PLA	polylactic acid
PLGA	poly(lactic-co-glycolic acid)
PLT	production logging tool
PT	pressure test
RBP	retrievable bridge plug
SP	straddle packer
SRT	step-rate test
WHP	wellhead pressure

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